```
#              -- Interview Questions ---                                   Date: 24-02-2026

# REV -- Another method
'''
r=["l","u","f","f","y"]
for i in range(len(r)):
for j in range(i,len(r)):
r[i], r[j] = r[j], r[i]
print(r)
'''
# Sort -- without using method

'''
s=[20,40,30,10]
#s=[5,6,4,2]
for i in range(len(s)):
for j in range(i,len(s)):
#   if s[i] < s[j]:                          # OP-- [40, 30, 20, 10]
if s[i] > s[j]:                          # OP-- [10, 20, 30, 40]
s[i],s[j] = s[j],s[i]
print(s)
'''
# add two lists:

l=[[1,2,3,],[10,20,30]]
k=[]
'''
for i in l:
k.extend(i)
print(k)
'''
# Another method:
m=[]
'''
for i in l:
for j in i:
m.append(j)
print(m)
'''

#vowels print in list:

'''
a="sanji"
b=[]
for i in a:
if i in "aeiou":
b.append(i)
print(b)
'''
#   op :  --  [[1, 4], [2, 5], [3, 6]]
'''
a=[[1,2,3],[4,5,6]]
b=[]
```

```
for i in range(len(a[0])):
b.append([a[0][i],a[1][i]])
print(b)
'''
# op: -- [[1,2,3],[4,5,6]]

'''
b=[[1, 4], [2, 5], [3, 6]]
c=[]
for j in range(len(b[0])):
c.append([b[0][j],b[1][j],b[2][j]])
print(c)
'''
# Remove Duplicate values:
s=[1,2,3,4,5,6,2,3]
t=[]
'''
for i in s:
if i not in t:
t.append(i)
print(t)
'''
# maximum without builtin methods:

##m=[2,4,6,34,75,98,12]
##n=a[0]
'''
for i in m:
if i>n:         # -- Maximum number
n=i
print(n)
'''
'''
for i in m:
if i<n:                 # -- Smallest number
n=i
print(n)
'''

#### ------   Communnication skills in CS file.txt                     ------------




## Assessment  ---  Second Largest Number
'''
a=[10,50,20,30,40]
b=0
c=0
for i in a:
if i>b:
b=i
print(b, "is the maximum value")
```

```
for i in a:
if i>c and i<b:
c=i
print(c, "is the second max value")
'''
#
```

```
# Second Largest Number                                   Date: 25-02-2026

a=[10,42,23,34,25]
l=a[0]
s=a[1]
'''
for i in range (len(a)):
if l<a[i]:
s=l
l=a[i]
elif s<a[i] and l!=a[i]:
s=a[i]
print(s)
'''
## Anangram
'''
a="heart"
b="earth"

print(sorted(a))
print(sorted(b))

if sorted(a) == sorted(b):
print("Anagram")
else:
print("Not Anangram")
'''

##  if a= [0,1,2,0,3]                  op: -- [1,2,3,0,0]
a=[0,1,2,0,3]
b=[]
c=[]
'''
for i in range(len(a)):
if a[i] != 0:
b.append(a[i])
else:
c.append(a[i])
print(b+c)
'''
'''
for i in a:
if i != 0:
b.append(i)
else:
c.append(i)
print(b+c)
```

```
'''
a=[0,1,2,0,3]
x=0              # --- Another Method
'''
for i in range(len(a)):
if a[i]!=0:
a[i],a[x]=a[x]i
print(a)
'''
# which two numbers addition gives 9
a=[0,2,3,4,6,7,11,9,5]           # --- op:[2,7]  two pointer means l and r
a.sort()
b=9
'''
l=0
r=len(a)-1
while l<r:
x=a[l]+a[r]
if x==b:
print([a[l],a[r]])
l+=1
#break
elif x<b:
l+=1
else:
r-=1
'''
# Intersection
a=[1,2,3,4]
b=[3,4,5,6,7]
c=[]
'''
for i in a:
for j in b:
if i==j and i!=c:
c.append(i)
print(c)
'''
# Alternate method in single loop: time less
'''
for i in a:
if i in b:
c.append(i)
print(c)
'''

#-------------------------------------------------------------------
# Set Methods without built-in methods

#  0p: --- {1,2,3,4,5,6} union
'''
a={1,2,3}
b={4,5,6}
for i in a:
```

```
b.add(i)
print(b)
'''
# intersection

a={1,2,3}
b={3,4,5}
'''
c=set()
for i in a:
if i in b:
c.add(i)
print(c)
'''
# difference
'''
for i in b:
if i in a:
a.remove(i)
print(a)
'''
#symmetric_difference
'''
c=set()

for i in a:
if i  not in b:
c.add(i)
for i in b:
if i not in a:
c.add(i)
print(c)
'''

# isdisjoint()
'''
a={1,2,3}
b={4,5,6}

print(a.isdisjoint(b))
print(b.isdisjoint(a))
'''
'''
a={1,2,3}
b={3,4,5,6}
for i in b:
if i in a:
a.remove(i)
print(a)
'''
#________________________________________________________________________________
```

Date: 26-02-2026

```
#Dictionary   ---  op; {1:1, 2:4, 3:27. 4:256}
'''
a=[1,2,3,4]
b={}
c=0

for i in a:
c=i**i
b[i]=c
print(b)
'''
#      op:       -- {'apple': 5, 'orange': 6, 'mango': 5, 'banana': 6}
'''
a=['apple', 'orange','mango','banana']
b={}
for i in a:
b[i]=len(i)
print(b)
'''

# Op: -- {'b': 1, 'l': 2, 'a': 2, 'c': 1, 'k': 1, ' ': 2, 'e': 1, 'g': 1, 's': 1, 'n': 1, 'j': 1, 'i': 1}

a="black leg sanji"
x={}

'''
count=0
for i in a:
if i in x:
x[i]+=1
else:
x[i]=1
print(x)
'''
### another method but have duplicate keys --- but it works in dictionary
count=0
'''
for i in a:
x[i]=a.count(i)
print(x)
'''
#  op:  -- {1:'a', 2:'b',3:'c', 4:'d'}
a={'a':1, 'b':2,'c':3,'d':4}
b={}
x=1
##c=a.keys()
##print(c)
'''
for i in a:
b[x]=i
x+=1
print(b)
'''
# another metjod
```

```
'''
for i in a.keys():
for j in a.values():
b[j]=i
print(b)
'''
# another method
'''
for i ,j in a.items():
b[j]=i
print(b)
'''
### y.update(x) with out built-in methods.     --  {.union() }

x={1:'a', 2:'b',3:'c', 4:'d'}
y={5:'e', 6:'f',7:'g', 8:'h'}
'''
for i,j in y.items():
x[i]=j
print(x)
'''
# .intersction() with out built-in methods.

x={'a':1,'b':2,'c':3,'d':4}
y={'c':6,'d':4,'e':5}
z={}
'''
for i,j in x.items():
for k,l in y.items():
if i==k and j==l:
z[i]=j
print(z)
'''
##### Important for key and value   i in y === key
'''
for i,j in x.items():
if i in y and y[i]==j:                ## -- y[i]==j
z[i]=j
print(z)
'''
# a.difference(b) with out built-in methods.

##a={'a':1,'b':2,'c':3}
##b={'c':3,'d':4,'e':5}
'''
for i,j in b.items():
if i in a and b[i]==j:
del a [i]
print(a)
'''
#b.differnce(a)
'''
for i,j in a.items():
if i in b and a[i]==j:
```

```
del b [i]
print(b)
'''
#a.symmetric-difference()

##a={'a':1,'b':2,'c':3}
##b={'c':3,'d':4,'e':5}
##
##for i,j in b:
##    if

#____________________________________________________________________________________
```

```
#      ## List comprehension                                    Date: 27-02-2026



##a=int(input())
##b=[]

# normal way
'''
for i in range(1,a+1):
b.append(i)
print(b)
'''
# sinle line
'''
c=[i for i in range(1,a+1) ]
print(c)
'''
# normal way of even numbers print:
'''
for i in range(1, a+1):
if i % 2 == 0:
b.append(i)
print(b)
'''
# single line even numbers using if andfor
'''
d=[i for i in range(1,a+1) if i%2 == 0]
print(d)
'''
# normal way of odd or even

'''
e=[]
for i in range(1,a+1):
if i%2==0:
e.append(f"{i} even")
else:
e.append(f"{i} odd")
print(e)
'''
```

```
# sinle line odd or even  using if,elsecand for

##f=[f" {i} even" if i%2 == 0 else f" {i} odd" for i in range(1,a+1)]
##print(f)

#___________________________________________________________________________
___________________________________________________________________________


## Nested for loop and if condition:

# check divisible by 3 and 5:

##a=int(input())
##b=[]

'''
for i in range(1,a+1):
if i%3 == 0:
if i%5 == 0:
b.append(i)
print(b)
'''
# single line
'''
c=[i for i in range(1,a+1) if i%3 == 0 if i%5 == 0]
print(c)
'''
# nested list add

x=[[10,20,30],[40,50,60]]
y=[]
'''
for i in x:
for j in i:
y.append(j)
print(y)
'''
# single line
'''
z=[j for i in x for j in i]
print(z)
'''
# Duplicate value remove:

x=[10,20,10,30,20]
y=[]

#      can't use in single line
'''
for i in x:
if i not in y:
y.append(i)
```

```
print(y)
'''
'''
for i in range(len(x)):
if x[0:i+1].count(x[i])==1:
y.append(x[i])
print(y)
'''
# single line:

##z=[x[i] for i in range(len(x)) if x[0:i+1].count(x[i])==1]
##print(z)

#___________________________________________________________________________
___________________________________________________________________________

### Set Comprehension:

##a=int(input())
##b=set()

# normal way
'''
for i in range(1,a+1):
b.add(i)
print(b)
'''
# sinle line
'''
c={i for i in range(1,a+1) }
print(c)
'''
# normal way of even numbers print:
'''
for i in range(1, a+1):
if i % 2 == 0:
b.add(i)
print(b)
'''
# single line even numbers using if andfor
'''
d={i for i in range(1,a+1) if i%2 == 0}
print(d)
'''
# normal way of odd or even

'''
e=set()
for i in range(1,a+1):
if i%2==0:
e.add(f"{i} even")
else:
e.add(f"{i} odd")
print(e)
```

```
'''

# sinle line odd or even  using if,elsecand for

##f={f" {i} even" if i%2 == 0 else f" {i} odd" for i in range(1,a+1)}
##print(f)

#____________________________________________________________________________________________
____________________________________________________________________________________________

## Nested for loop and if condition:

# check divisible by 3 and 5:

##a=int(input())
##b=set()

'''
for i in range(a):
if i%3 == 0:
if i%5 == 0:
b.add(i)
print(b)
'''
# single line
'''
c={i for i in range(a) if i%3 == 0 if i%5 == 0}
print(c)
'''
# nested list add

x={10,20,30},{40,50,60}
y=set()
'''
for i in x:
for j in i:
y.add(j)
print(y)
'''
# single line
'''
z={j for i in x for j in i}
print(z)
'''
#____________________________________________________________________________________________
____________________________________________________________________________________________

### Dictionary Comprehension:

##a=int(input())
##b={}

# int power of 2
'''
```

```
for i in range(1,a+1):
b[i]=i**2
print(b)
'''
# singleline
'''
c={i:i**2 for i in range(1,a+1)}
print(c)
'''
# int power of 2 if int is even num:
'''
for i in range(1,a+1):
if i%2 == 0:
b[i]=i**2
print(b)
'''
#sinle line:

'''
d={i:i**2 for i in range(1,a+1) if i%2 == 0  }
print(b)
'''
#
'''
for i in range(1,a+1):
if i%2==0:
b[i]="even"
else:
b[i]="odd"
print(b)

e={i:"even" if i%2==0 else "odd" for i in range(1,a+1)}
print(e)
'''

#
# 	Saturday -- mock test for python data structures		Date: 28-02-2026

#
#	Sunday -- holiday					Date: 01-03-2026


#


# Gnerator Comprehension		Date: 05-03-2026
```

```
a=5
'''
d=(i for i in range(a))

print(next(d))
print(next(d))
print(next(d))
print(next(d))
print(next(d))
print(next(d))     # error -- StopIteration

'''
# even numbers:
'''
b=(i for i in range(1,a+1) if i%2 ==0)
print(next(b))
print(next(b))
'''
# if and else odd or even:

'''
c=( f"{i} odd" if i%2==1   else f"{i} even" for i in range(1,a+1) )
for i in range(a):
print(next(c))
'''



a=[40,20,30,40,50]
lar = a[0]
secl = a[1]

for i in range(len(a)):
if a[i]>lar:
secl=lar
lar=a[i]
elif secl < a[i] and lar!=a[i]:
secl=a[i]
print(lar,secl)
```